

## Chapter 1 Foundations of Statistical Modelling Demonstrated with Simple Regression

### Chapter Overview

The major objective of this chapter is to develop an understanding of the principles of statistical modeling in general and the simple linear regression model in particular. These principles provide a conceptual foundation for the remainder of the text. The main topics of this chapter include:

- Definition and description of statistical modelling as a guiding theme for the text
- Foundational principles for simple linear regression
- Specification and estimation of the simple linear regression model
- Statistical inference with the simple linear regression model
- Binary variables in simple linear regression
- Basic regression diagnostics for simple linear regression

### Simple Regression Models

The statistical methods covered in this text all involve estimating and interpreting parametric models. A parametric model is a mathematical expression that uses *parameters* to represent hypothetical relations among variables in the population. In other words, the model represents a hypothetical process that generated the outcome variable(s) according to certain population parameters. It is likely that you are already familiar with simple regression, which is usually covered in introductory textbooks and courses. Here, we study simple regression in detail to establish a foundation for presenting more complex modelling procedures because the same principles and methods continue to apply as the simple regression model is expanded into larger models for more than two variables. Thus, the first model we will examine extensively is the **simple linear regression model** in which one variable, generically referred to as  $Y$ , is a linear function of only one other variable, generically referred to as  $X$ .

To begin nailing down the terminology and notation used in this text, let's take a quick, initial look at the simple linear regression model. It's a model for the numerical score of case  $i$ , or individual  $i$ , on some *outcome variable*  $Y$  given that individual's score on some *regressor*  $X$ . "Individual  $i$ " is a potentially observed unit in the population of interest; in social science research, these units of observation are often individual people (i.e., research participants), but in other applications, the units of observation might instead be animals, municipalities, parent-child dyads, or business firms, among other possibilities. One way to write the simple regression model is with the equation

$$(1.1) \quad Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i.$$

A model that describes (and possibly explains or predicts) population-level individual differences in  $Y$  is characterized by its parameters; that is, a **parameter** is a characteristic of a population; this text follows the standard notational practice in the statistics and methodology literature of using Greek letters to represent population parameters. The parameters in the simple regression model are the **intercept**, denoted  $\beta_0$ , and the **slope**, denoted  $\beta_1$ . The slope term is also often called the **regression coefficient** because in the language of basic algebra,  $\beta_1$  is the coefficient of variable  $X_i$  in the equation. There are *three* variables in this simple regression

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model in that there are three terms with the subscript  $i$ , indicating that they vary across cases (or units of observation). The parameters do not have the  $i$  subscript because they are *constants*; they do not vary across cases.

The **outcome** variable is  $Y_i$ ; terms that are essentially synonymous with “outcome variable” include *dependent variable*, *response*, and *criterion*. The **regressor** is  $X_i$ ; terms that are essentially synonymous with “regressor,” depending on context, include *independent variable*, *predictor*, *explanatory variable*, and *covariate*. I have described these terms as essentially synonymous because they are treated the same way mathematically when statistical models are fitted to data. But if the purpose of a statistical model is formal causal inference, then the terms *independent variable* or *explanatory variable* are appropriate synonyms for regressor, whereas if the purpose is statistical description or prediction, then the terms *predictor* or *covariate* are appropriate synonyms for regressor. Some researchers prefer to use the term *independent variable* only to refer to an experimentally manipulated variable.

The third variable in Equation 1.1 is the **error** term,  $\varepsilon_i$ , which arises to represent the inaccuracy of the model’s ability to perfectly reproduce the value of the outcome variable for a given observation. Unlike  $Y_i$  and  $X_i$ , which are directly measured by the researcher, the error is an unobserved variable which is not directly measured and instead arises as a property of the model. Although it is not explicitly indicated in the Equation 1.1, the simple regression model also includes a parameter,  $\sigma^2$ , to capture the variance of the errors, that is,  $\sigma^2 = \text{VAR}(\varepsilon_i)$ .

Estimates of model parameters calculated from sample data are statistics, but I also refer to them as **parameter estimates** throughout this text. To distinguish parameter estimates from the population parameter, I use the “hat” symbol. For example,  $\hat{\mu}$ , or “mu hat,” is the sample estimate of the mean parameter  $\mu$  (also commonly denoted as  $\bar{Y}$  to represent the sample mean of variable  $Y$ ), while  $\hat{\beta}$ , or “beta hat,” is the estimate of a regression slope parameter  $\beta$ .

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### Section recap: Simple linear regression model

The simple linear regression model for the relation between an outcome variable  $Y$  and a predictor  $X$  is

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i.$$

The **intercept** parameter,  $\beta_0$ , is the predicted value of  $Y$  when  $X$  equals 0.

The **slope** parameter,  $\beta_1$ , is the predicted difference in  $Y$  per one-unit higher value on  $X$ .

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### Research example

To provide a context for using the simple regression model, let’s consider an actual dataset: A researcher is interested in whether movies with better reviews from critics earn a larger profit for the filmmakers. The researcher obtains data for  $N = 609$  movies released between 2008 and 2012, including their net *profit* (measured as the difference between their gross income and budget in millions of American dollars) as well as a numerical score giving the percentage of professional critical reviews that are positive for a given film. Thus, individual

movies comprise the units of observation for this example. Simply put, the researcher wants to devise a model for *profit* (measured in millions of American dollars) to describe (or explain or predict) how movies vary on this important outcome. This dataset is available on the text's webpage along with annotated input and output showing how to reproduce the analyses presented in this chapter.

Before fitting a model to empirical data, it is always wise to examine the data using descriptive statistics and graphs. Thus, we begin by looking at the sample distribution of the movie *profit* outcome variable. Because *profit* can be considered a continuous variable, we can examine its distribution graphically with the boxplot and histogram in Figure 1.1. These plots show that the sample distribution of *profit* ranges from positive to negative values (some movies lost a lot of money!), with the centre of the distribution very near 0. The distribution is unimodal with a slight positive skewness, but its most notable feature is that there is a very high frequency of observations piled up around a relatively narrow range between -50 and 50, indicating that the distribution has high kurtosis. Here, I am intentionally avoiding a comparison of this sample distribution with a hypothetical normal population distribution; the purpose is simply to describe the distribution shape to help understand the data. Whether the shape is consistent with a normal distribution is irrelevant to our eventual use of the linear regression model because *ordinary linear regression does not assume that the observed outcome variable is normally distributed*. Instead, as I explain in detail later, a normal distribution assumption applies to the model's errors.

[Figure 1.1 here]

Next, we might wish to have some numerical summaries, or descriptive statistics, to describe the sample distribution of *profit*. Table 1.1 provides such statistics for *profit* as well as several other variables in the movie dataset. These summary statistics support the observations made from the graphs in Figure 1.1, particularly that the *profit* distribution is positively skewed (*skewness* = 2.59) and the large kurtosis value (*kurtosis* = 13.94) arises because there are many movies with *profit* values within a relatively small range (25<sup>th</sup> percentile = -12.84; 75<sup>th</sup> percentile = 32.52) despite that the entire distribution ranges from a minimum *profit* of -176.92 to a maximum of 523.51.

[Table 1.1 here]

### Developing the simple regression model

In the movie example, the main goal is to model *profit* as a function of critical reviews as captured by the *review* variable (as described earlier, the percentage of positive critical reviews for a given film). Before specifying and estimating the model, we should graph the data to get a sense of the bivariate association between *review* and *profit*. Specifically, we can use a scatterplot to visualize the distribution of *profit*, the outcome variable, *conditioned on* the regressor variable, *review*. This scatterplot, shown in Figure 1.2, suggests that there is not a particularly strong association between *review* and *profit*: Most movies have a *profit* between -100 and 100 regardless of their *review* score. Yet, the movies with the highest *profit* seem to have relatively high *review* scores (e.g., a *profit* greater than 100 only occurs for movies with a *review* score of at least 20), and the movies with the highest profit do tend to have high *review*

scores, although there are also movies with very high *review* scores that have a negative *profit*. Next, we summarize these observations about the scatterplot using a descriptive statistic, namely the *product-moment correlation*, before ultimately presenting the linear regression model.

[Figure 1.2 here]

The sample **product-moment covariance** between two variables  $X$  and  $Y$  is the basic ingredient that allows us to estimate correlations as well as the parameters of linear models (including complex, multivariate models). Recall that the sample *variance* of a variable  $Y$  is calculated based on the deviation from the mean, or  $Y_i - \bar{Y}$ , for each case  $i$ :

$$(1.2) \quad \text{VAR}(Y) = s_Y^2 = \frac{\sum_{i=1}^N (Y_i - \bar{Y})^2}{N-1} = \frac{\sum_{i=1}^N (Y_i - \bar{Y})(Y_i - \bar{Y})}{N-1} = \frac{SS_Y}{N-1},$$

where  $SS$  stands for *sum-of-squares*. Variance quantifies the spread of the distribution (i.e., the average squared deviation from the mean), and the square root of variance leads to the familiar standard deviation statistic:  $SD(Y) = s_Y = \sqrt{\text{VAR}(Y)}$ .

The formula for sample covariance is very similar to the variance formula, but it incorporates both  $Y$  and  $X$ :

$$\text{COV}(Y, X) = s_{YX} = \frac{\sum_{i=1}^N (Y_i - \bar{Y})(X_i - \bar{X})}{N-1} = \frac{SCP}{N-1},$$

where  $SCP$  stands for *sum of cross-products* in that  $(Y_i - \bar{Y})(X_i - \bar{X})$  is the *cross-product* between  $Y$  and  $X$  for case  $i$ . Thus, whereas variance quantifies how much  $Y$  varies with itself univariately, covariance quantifies the amount that  $Y$  covaries with  $X$  bivariate. In the movie example, the covariance between *profit* and *review* is  $s_{YX} = 467.98$ . But covariance depends on the scales of  $Y$  and  $X$ ; for this reason, a covariance value can be difficult to interpret as a basic descriptive statistic measuring the (linear) association between  $X$  and  $Y$ .

Therefore, a covariance can be standardized into a **Pearson product-moment correlation** (usually simply referred to as “correlation” or “correlation coefficient”) for interpretation. The population correlation is represented with the Greek letter *rho*,  $\rho$ , while the sample correlation is represented with the Roman letter  $r$ . Thus,  $r$  estimates  $\rho$ . As readers are likely aware, a correlation is a value between -1 and +1. Equation 1.3 shows that the correlation between  $X$  and  $Y$  is based on the covariance between  $X$  and  $Y$  and their variances:

$$(1.3) \quad r = \frac{\text{COV}(Y, X)}{\sqrt{\text{VAR}(Y) \text{VAR}(X)}} = \frac{s_{YX}}{\sqrt{s_Y^2 s_X^2}}.$$

In the movie example, the correlation between *review* and *profit* is  $r = .30$ , suggesting that there is a tendency for movies with better reviews to earn a greater profit, although this correlation is rather modest, consistent with our observations from Figure 1.2. A 95% confidence interval estimate of  $\rho$  is (.23, .37), that is, with 95% confidence, the interval from .23 to .37 captures the population correlation; that is, the interval from .23 to .37 contains a range of plausible values for the population correlation between *review* and *profit*.

A correlation describes the extent to which two variables are *linearly* associated, implying that a straight line going through the middle of the points in the scatterplot does an adequate job of representing, or *modeling*, the association between  $X$  and  $Y$ . It is easier to assess the appropriateness of the straight-line model if we add a *non-parametric* LOWESS (“LOcally WEighted Scatterplot Smoothing”) curve to the plot, which is designed to capture subtle, nonlinear regularities in the data that might not be well represented with the parametric

regression line (see Fox, 2016, for details on calculation of LOWESS curves).<sup>1</sup> As shown in Figure 1.3 for the movie example, the LOWESS curve is somewhat consistent with the straight line, despite that they are not perfectly aligned. Hence, although the relation between *review* and *profit* is not particularly strong, it does appear that the ordinary simple regression line is a reasonable *model* for the relation. That is, there doesn't seem to be any strong pattern in the data that would be misrepresented by a straight line that best fits the data.

[Figure 1.3 here]

Before considering the mathematical formulas for the linear model, it is helpful to distinguish between **predicted values** of  $Y$ , also known as *fitted* or *model-implied* values, and **observed values** of  $Y$ . The observed values of  $Y$  are the actual measured values for the outcome that are in the dataset. In the movie example, there are  $N = 609$  cases (individual movies) with *profit* values. Each of these 609 *profit* scores is an observed value of  $Y$ . Next, Figure 1.3 shows that most of the 609 datapoints do not fall on the regression line, and some are rather far from the line. The predicted values of  $Y$ , represented as  $\hat{Y}$ , are the  $Y$  values determined by the regression line across the  $X$  continuum. As we are using *review* scores to predict or model *profit* scores, then the  $\hat{Y}$  values are the predicted, model-implied *profit* values at each possible *review* score. For example, consider a movie with *review* score = 0: The height of the regression line in Figure 1.3 directly above *review* = 0 corresponds to a *profit* slightly below 0, or -15.60 to be exact. The figure shows two datapoints above *review* = 0; for both,  $\hat{Y} = -15.60$ , but one has an observed *profit* very close to  $\hat{Y}$  (the datapoint is almost directly on the line) whereas the other has an observed *profit* very slightly above 0.

Because the correlation between *review* and *profit* doesn't equal 1, most of the observed *profit* scores don't equal the corresponding predicted value from the linear regression of *profit* on *review*. The **residual** for an individual case  $i$  is the difference between that case's observed  $Y_i$  value and the corresponding  $\hat{Y}_i$  value based on the estimated regression line:

$$e_i = Y_i - \hat{Y}_i$$

Thus, datapoints falling exactly on an estimated regression line have residuals equal to zero, whereas datapoints that are far from the line have large residuals; the residual for case  $i$  equals the vertical distance between that case's datapoint and the regression line fitted to the scatterplot. The residual,  $e_i$ , is not the same as the error,  $\varepsilon_i$ , presented earlier in Equation 1.1. When  $\hat{Y}_i$  is calculated based on a regression line *estimated* from sample data, then the difference  $Y_i - \hat{Y}_i$  is the residual for case  $i$ . Conversely, when  $\hat{Y}_i$  is based on (unknown) population parameters for a regression line (as in Equation 1.1), then the difference  $Y_i - \hat{Y}_i$  is the error for case  $i$ .

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### Section recap: Developing the simple linear regression model

<sup>1</sup> A LOWESS curve is susceptible to chance variations in data with small sample sizes ( $N < 100$  or so), such that only a few cases can cause the curve to display dramatic bends. In such a situation, the curve may *overfit* the data rather than smoothing over minor, sample-specific variation. In the movie example, the sample size is rather large, and so the LOWESS curve is resistant to the influence of a few unusual cases.

A scatterplot of an outcome variable against a potential regressor helps determine whether it is appropriate to *model* the relation between the two variables using a straight-line function, i.e., a linear regression equation.

**Covariance** is a descriptive statistic measuring the strength of linear association between two variables; **product-moment correlation** is a covariance which has been standardized to range between -1 and +1.

A **predicted value**,  $\hat{Y}_i$ , is the outcome variable score for case  $i$  that is *predicted* from the regression line given that case's score,  $X_i$ , on the regressor variable,.

An **observed value**,  $Y_i$ , is the actual outcome variable score for case  $i$  regardless of that case's score on the regressor.

Given a population regression equation, the regression error for individual  $i$  is  $\varepsilon_i = Y_i - \hat{Y}_i$ .  
Given an *estimated* regression equation, the **residual** for individual  $i$  is  $e_i = Y_i - \hat{Y}_i$ .

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### Simple linear regression: Model specification

**Model specification** refers to establishing a model's parameters with one or more equations giving the hypothetical mathematical relations among the variables in terms of population parameters. Often specification also involves statements about a model's assumptions regarding variance terms or probability distribution, although these assumptions sometimes arise because of estimation method (see below) and are not always a part of specification. Specification of the simple linear regression model was presented earlier, but I reiterate it here with more detail.

The one-predictor *linear regression model* can be specified in two equivalent ways. The first expression is a function of the predicted values of the outcome variable:

$$(1.4) \quad \hat{Y}_i = \beta_0 + \beta_1 X_i,$$

where  $\hat{Y}_i$  is the predicted outcome value for case  $i$  and  $X_i$  is the observed value of the regressor for the same case  $i$ . Next,  $\beta_0$  is the **intercept** parameter, giving the value of  $\hat{Y}$  corresponding to  $X = 0$ . Then  $\beta_1$  is the **slope** parameter, also called the **regression coefficient**, which is the amount that  $\hat{Y}$  differs when comparing two hypothetical cases that differ by one unit on  $X$ . In other words, a one-unit higher score in  $X$  is associated with a difference in  $\hat{Y}$  equal to  $\beta_1$ . Of course, sometimes  $\beta_1$  is a negative number, indicating the amount that  $\hat{Y}$  decreases per unit difference in  $X$ , just as a correlation can be positive or negative.

The second way of expressing the same model substitutes the observed  $Y$  for  $\hat{Y}$  on the left-hand side of the equation:

$$(1.5) \quad Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i,$$

where  $Y_i$  is the observed outcome score for case  $i$ ,  $\beta_0$ ,  $\beta_1$ , and  $X_i$  are the same as above, and  $\varepsilon_i$  is the regression error for case  $i$ . Because  $\varepsilon_i = Y_i - \hat{Y}_i$ , it is easy to see that the two expressions of the model (Equations 1.4 and 1.5) are algebraically equivalent. Finally, the error variance,  $\sigma^2 =$

$\text{VAR}(\varepsilon_i)$ , is the model's last parameter, which summarizes the extent to which observed values differ from predicted values across the population.

Because  $\beta_1$  represents the strength of the association between the regressor variable and the outcome, it is an **effect size** parameter. Specifically, because  $\beta_1$  describes the linear association between  $X$  on  $Y$  in terms of the scale of  $Y$ , it is an *unstandardized effect size* measure. The correlation is a type of *standardized effect size* measure because it indicates the strength of the linear relation between two variables on a standard scale from -1 to +1 rather than the observed scale of the outcome variable.

### Simple linear regression: Model estimation

**Model estimation** refers to the process of using sample data to calculate estimates of a model's parameters. For any model, there may be several potential methods of estimation, but of course some are better than others; parameter estimates from an optimal estimation method are *unbiased*, *consistent*, and *efficient*, provided the method's assumptions are met. Briefly, a parameter estimate is unbiased if the mean of its sampling distribution equals the true value of the parameter at a given sample size; an estimate is consistent if its value approaches the parameter value as the sample size increases toward infinity; and an estimate is efficient if, compared to other estimation methods, its sampling distribution has the smallest variance.

In simple linear regression, the parameters  $\beta_0$  and  $\beta_1$  are almost always estimated according to the **ordinary least-squares (OLS)** method. When its assumptions are met, parameter estimates calculated with OLS are unbiased, consistent, and maximally efficient. These assumptions are addressed in this chapter and subsequent chapters.

The OLS method gives formulas for the intercept and slope parameter estimates,  $\beta_0$  and  $\beta_1$ , such that the squared residuals in the sample are as small as possible. Specifically, the regression line defined by the OLS estimates  $\beta_0$  and  $\beta_1$  minimizes the **sum of squared residuals**:

$$SS_e = \sum_{i=1}^N e_i^2 = \sum_{i=1}^N (Y_i - \hat{Y}_i)^2.$$

Substituting the simple regression line as the model for  $\hat{Y}$ , we see that the residual sum of squares depends on the values for the estimates of  $\beta_0$  and  $\beta_1$ :

$$SS_e = \sum_{i=1}^N (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^N (Y_i - [\beta_0 + \beta_1 X_i])^2.$$

Thus, the goal is to find values for  $\beta_0$  and  $\beta_1$  that make  $SS_e$  as small as possible; calculus is required to derive the formulas for  $\beta_0$  and  $\beta_1$  that do so. The resulting formula for the OLS estimate of  $\beta_1$  is

$$(1.6) \quad \beta_1 = \frac{\text{COV}(Y, X)}{\text{VAR}(X)} = r \left( \frac{s_Y}{s_X} \right).$$

With this expression, it is easy to see that if both  $X$  and  $Y$  are *standardized* (i.e., transformed so that their variances equal 1), then the slope estimate equals the correlation between  $X$  and  $Y$ .<sup>2</sup>

Next, given the estimate of  $\beta_1$ , the OLS estimate of  $\beta_0$  is:

$$\beta_0 = \bar{Y} - \beta_1 \bar{X}.$$

For the movie example modelling *profit* as a linear function of *review*, the parameter estimates are  $\beta_0 = -15.61$  and  $\beta_1 = 0.70$ . Thus, the estimated OLS regression line is:

$$\hat{Y}_i = -15.61 + 0.70X_i$$

<sup>2</sup> Standardized regression coefficients are discussed in Chapter 2.

or

$$\text{profit}_i = -15.61 + 0.70\text{review}_i.$$

This is the equation of the straight line going through the cloud of datapoints in Figure 1.3. In fact, an OLS regression line is guaranteed to pass through the datapoint  $(\bar{X}, \bar{Y})$ . The slope estimate  $\beta_1 = 0.70$  indicates that a one-unit (i.e., one-percent) higher *review* score predicts a \$0.70 million greater *profit*.

Often, the scale of the regressor  $X$  is such that a one-unit difference is quite small, making it difficult to gauge the meaningfulness of a slope estimate. In the movie example, a one-unit difference on *review* corresponds to only one percent. But with linear models, we can multiply a slope estimate by 10 so that it indicates the predicted difference in  $Y$  per 10-unit difference in  $X$  instead of a one-unit difference. Thus, for the movie example,  $\beta_1 = 0.70$  implies that \$7 million (i.e.,  $10 \times 0.70$ ) is the predicted *profit* difference per 10% higher *review* score.

Next, the intercept estimate  $\beta_0 = -15.61$  indicates that a movie with a *review* score equal to zero is predicted to lose \$15.61 million. In many applications of linear regression, the intercept parameter is not substantively useful because  $X = 0$  is outside of the range of the data (e.g., perhaps because of the way a questionnaire scale is scored); here, *review* = 0 is possible (in fact, there are two movies in the sample with *review* = 0) and so the intercept estimate is meaningful. But even when  $X = 0$  is a possible measured value, the intercept is usually much less important than the slope.

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### Section recap: Specification and estimation of the simple linear regression model

**Specification** establishes a model's parameters using one or more equations for the hypothetical relations among the variables. Specification may also involve assumptions about a model's variance terms or probability distributions.

Once a model is specified, **estimation** is the procedure by which the parameters are estimated from sample data.

Ordinary least squares (OLS) is the most common estimation method for linear regression. OLS produces parameter estimates that minimize the sum of squared residuals.

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### Simple linear regression: Inference

Consistent with the weak association between *review* and *profit* observed in the scatterplot, the slope estimate  $\beta_1 = 0.70$  does not seem like a strong effect given the observed scales of *review* and *profit*. Thus, an important aspect of model interpretation typically involves significance testing and confidence interval estimation to infer whether this slope estimate is distinguishable from a population slope equaling zero (using a significance test) or, more comprehensively, to determine a plausible range for the value of the population slope (using a confidence interval).



To reach inferences about the unknown population parameters based on the OLS parameter estimates, it is necessary to make some assumptions about the error random variable,  $\varepsilon_i$ , in Equation 1.1; we can eventually assess these assumptions using the residuals from the estimated model. Primarily, we assume that  $\varepsilon_i \sim N(0, \sigma^2)$ , meaning that the errors are normally distributed with mean equal to zero and variance equaling  $\sigma^2$ . Thus, contrary to popular perception, in OLS regression, a normal distribution assumption pertains to the unobserved errors and there is *no* assumption that either of the measured variables  $Y$  or  $X$  is normal. Additionally, the fact that there is no  $i$  subscript on  $\sigma^2$  reflects the assumption that the error variance is the same (i.e., **constant variance**) for all observations, regardless of their value for  $X$ . This constant variance assumption is also known as **homogeneity of variance** or **homoscedasticity**: The variance of the errors is assumed homogeneous, or constant, across all values of  $X$ . Finally, we also assume that the errors are independent; we address this issue in a later chapter on multilevel modeling.

Researchers are usually interested in testing the null hypothesis that the slope parameter equals zero:

$$H_0: \beta_1 = 0.$$

The ratio of OLS slope estimate to its estimated **standard error** follows a  $t$ -distribution with  $(N - 2)$  degrees of freedom. Thus, the null hypothesis for the slope is evaluated with a  $t$ -test:

$$(1.7) \quad t = \frac{\beta_1}{s_{\beta_1}},$$

where  $s_{\beta_1}$  is the estimated standard error of  $\beta_1$ . A formula for this standard error estimate is

$$s_{\beta_1} = \sqrt{\frac{SS_e/(N-2)}{SS_x}}.$$

Recall that the *standard error* of a statistic is the standard deviation of its sampling distribution; hence, standard error reflects the average amount that a statistic (such as a sample mean or a slope estimate) randomly drawn from its sampling distribution differs from the true parameter value.

Additionally, we can get a confidence interval estimate of  $\beta_1$  using the usual symmetric confidence interval approach based on the  $t$ -distribution:

$$\beta_1 \pm s_{\beta_1} \times t_\alpha,$$

where  $t_\alpha$  is the appropriate critical  $t$ -value for a  $(1 - \alpha)\%$  confidence interval and  $\alpha$  is the predetermined probability of a Type I error (usually  $\alpha = .05$ , leading to 95% confidence intervals). There is an exact correspondence between the  $t$ -test and the confidence interval in that the null hypothesis is retained if the confidence interval overlaps 0 and rejected if the confidence interval does not contain 0. Ultimately though, the confidence interval is more informative than the significance test: Not only does the confidence interval indicate whether 0 (the value given by the null hypothesis) is a plausible value for the population slope parameter, but also the confidence interval gives a whole range of plausible values for the parameter.

This form of the  $t$ -test and confidence interval construction also applies to the intercept estimate, but these results usually have no substantive interest (despite that statistical software typically includes significance test results for the intercept in default output).

In the movie example of the regression of *profit* on *review*,  $s_{\beta_1} = 0.09$ ; therefore

$$t = \frac{\beta_1}{s_{\beta_1}} = \frac{0.70}{0.09} = 7.75.$$

The two-tailed  $p$ -value for this  $t$ -statistic is less than .0001; thus, with Type I error probability  $\alpha = .05$ , we reject the null hypothesis that the population slope equals zero. Therefore, although the  $\beta_1 = 0.70$  is somewhat small, the large sample size ( $N = 609$ ) leads to a small standard error relative to  $\beta_1$ , which in turn leads to a conclusion that  $\beta_1$  is statistically significant, meaning that the sample provides adequate evidence to conclude that the true parameter,  $\beta_1$ , is unlikely to equal 0. Compared with a null hypothesis test, a confidence interval around the slope estimate provides more information regarding likely values of the population slope. For the movie example, the 95% confidence interval estimate of  $\beta_1$  is (0.53, 0.88), suggesting that any value between 0.31 and 0.65 is a plausible guess for the true value of  $\beta_1$ .

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### Section recap: Statistical inference with the simple linear regression model

The null hypothesis that  $\beta_1 = 0$  can be evaluated using a  $t$ -test with

$$t = \frac{\beta_1}{s_{\beta_1}}$$

where  $s_{\beta_1}$  is the estimated standard error of  $\beta_1$ .

More information about the true parameter is provided by a confidence interval, which can be calculated with

$$\beta_1 \pm s_{\beta_1} \times t_\alpha$$

where  $t_\alpha$  is the critical  $t$ -value for a  $(1 - \alpha)\%$  confidence interval with  $\alpha$  as the predetermined Type I error probability.

The validity of these inferential methods depends on the assumption that the errors are normally distributed with constant (homogenous) variance across the range of  $X$ .

[end box]

### Simple regression with a binary regressor

The simple regression model easily accommodates a binary regressor (i.e., a variable with only two values or categories, or *dichotomous*), although the outcome variable should still be continuous. For instance, continuing with the movie example, suppose we're interested in comparing the *profit* of movies released in 2008 (during a global recession) with the *profit* of movies released in 2009. If we limit the dataset to movies from only these two years, then *year of release* is a binary variable.

A scatterplot depicting the relation between *year* and *profit* is in Figure 1.4, with *year* represented by a **dummy variable** such that movies released in 2008 are assigned a 0 and movies from 2009 are assigned a 1.<sup>3</sup> Because *year* is a binary variable, the choice of numerical values for its two categories is arbitrary (indeed, we could have just stuck with the values 2008 and 2009), and the results presented here generalize to any numerical coding scheme that

<sup>3</sup> Dummy coding is presented in more detail in Chapter 3.

distinguishes between the two values of *year*. But dummy variable codes of 0 and 1 produce an especially convenient interpretation of the OLS regression parameters, as we will see below. In other applications, the values of a binary regressor might originally be represented with words or characters, but it is necessary to assign numerical codes to these categories to represent the variable in a regression model (e.g., *control* = 0; *treatment* = 1). Because there are only two values of the regressor, a straight line is the mathematically simplest way to connect the middle of the points in the plot above  $X = 0$  to the middle of the points above  $X = 1$ . Specifically, when  $X$  is binary and coded 0 or 1, a regression line will connect the point  $(0, \bar{Y}_{X=0})$  to the point  $(1, \bar{Y}_{X=1})$ . This line is superimposed in Figure 1.4 for the movie example.

[Figure 1.4 here]

With a binary regressor coded into two numerical values, the formulas already presented for OLS regression parameter estimates are directly applicable, with the two values used for  $X_i$ . In the movie example, the estimated regression line modelling *profit* as a function of *year* is

$$\hat{Y}_i = 13.55 + 9.23X_i$$

Because  $\beta_0$ , the estimated intercept, is  $\bar{Y}$  when  $X = 0$ , then 13.55 is the predicted *profit* when the dummy variable equals 0, which corresponds to movies released in 2008. The intercept estimate is in fact the mean *profit* for 2008 movies:

$$\bar{Y}_{X=0} = 13.55 + 9.23(0) = 13.55 = \bar{\text{profit}}_{2008}.$$

Next, because  $\beta_1$ , the estimated slope, is the difference in  $\bar{Y}$  when  $X_i$  differs by one, then 9.23 is the predicted *profit* difference between 2008 movies and 2009 movies. This slope estimate equals the *difference* between the 2008 mean and the 2009 mean:

$$\bar{Y}_{X=1} = 13.55 + 9.23(1) = 22.78 = \bar{\text{profit}}_{2009}.$$

This mean difference,  $\beta_1 = 9.23$ , is a very small effect given that the overall standard deviation of *profit* is 60.51, consistent with the flat slope of the regression line in Figure 1.4.

Regarding inference, when the sole regressor in a simple regression model is binary, the  $t$ -test for the slope (Equation 1.7) is equivalent to the well-known independent-groups  $t$ -test comparing the outcome variable means of the two groups formed by the binary regressor:

$$t = \frac{\beta_1}{s_{\beta_1}} = \frac{\bar{Y}_2 - \bar{Y}_1}{s_{\bar{Y}_2 - \bar{Y}_1}}$$

where  $s_{\bar{Y}_2 - \bar{Y}_1}$  is the standard error of the difference between means calculated using the familiar pooled-variance statistic formed under the homogeneity of variance assumption. Thus, the independent-groups  $t$ -test is a special case of significance testing in OLS regression. Likewise, the  $p$ -value for the correlation between a binary  $X$  and a continuous  $Y$  will equal that for the  $t$ -test.<sup>4</sup> In the current example, this  $t$ -test is not significant,  $t(246) = 1.15$ ,  $p = .25$ , implying that the population mean *profit* difference between 2008 movies and 2009 movies may not differ from zero. Finally, the confidence interval for the regression slope is identical to a confidence interval for the difference between two independent means; in the current example, the 95% confidence interval is  $(-6.63, 25.09)$ , conveying a range of plausible values for the population mean difference,  $\mu_{2009} - \mu_{2008}$ .

<sup>4</sup> The product-moment correlation between a binary variable and a continuous variable calculated according to Equation 1.3 is known as the *point-biserial* correlation. Here, the point-biserial correlation between *year* and *profit* is  $r = .07$ .

### Binary $Y$ ?

In contrast with applications where a continuous  $Y$  is regressed on a binary  $X$ , it is critical to recognize that when the outcome  $Y$  itself is binary, a linear regression model is not appropriate for the data. Instead, it is appropriate to model a binary outcome using a nonlinear modelling procedure such as logistic regression (see Chapter 6).

[start box]

#### Section recap: Binary variables in the simple linear regression model

When a continuous outcome variable is regressed on a binary dummy variable, the intercept equals the mean of the category coded zero,  $\beta_0 = Y_{X=0}$ , and the slope equals the difference between the means of the two categories,  $\beta_1 = (Y_{X=1} - Y_{X=0})$ .

Furthermore, the  $t$ -test for the slope estimate is equivalent to the independent-groups  $t$ -test and a  $(1 - \alpha)\%$  confidence interval for the slope is identical to a  $(1 - \alpha)\%$  confidence interval for the difference between the two means.

[end box]

### Basic Regression Diagnostic Concepts

**Regression diagnostics** are graphical and numeric methods for evaluating the extent to which a regression model adequately represents the data and for evaluating the trustworthiness of inferential conclusions about the model's parameter estimates. Regression diagnostics are used to check the extent to which the model's assumptions have been violated and to examine whether any unusual observations may be impacting the results. Although diagnostic methods mostly show their strength when applied to models with more than one regressor, I introduce the basic concepts here because they can be more simply illustrated with the simple regression model.

#### Linearity

First and foremost, the simple regression model specifies a *linear* relation between the predictor and outcome variables. Linearity is most easily examined with a scatterplot of the observed variables (or a scatterplot of residuals; see Chapter 2), which can be enhanced with a non-parametric regression curve as demonstrated in Figure 1.3. In the movie example, the straight-line regression model does seem to be an adequate representation of the positive (but weak) relation between *review* and *profit*. The online supplement to this chapter provides an example for which a straight-line model may not be appropriate.

#### Distribution of residuals

Next, correct inference for parameter estimates (i.e., significance tests and confidence intervals) assumes that the model's errors are normally distributed with constant, or homogeneous, variance,  $\varepsilon_i \sim N(0, \sigma^2)$ . If not, the Type I error rate (if the null hypothesis is true) and power (if the null hypothesis is false) of the  $t$ -test for the slope parameter estimate is compromised, and the widths of their confidence intervals are incorrect. Because of the central limit theorem, if the sample size is large ( $> 100$  or so for a simple regression model), then the

**Commented [DBF2]:** Regression of alcohol use on age

**Commented [DBF3R2]:** Online supplement to be discussed in chapter 0

normality assumption is less critical and inference is still valid, as long as the other assumptions are met. However, nonnormality still impacts the *efficiency* of OLS estimates, meaning that alternative estimators of the model may lead to greater statistical power and narrow confidence intervals for population inference. Therefore, it is important to evaluate the tenability of the assumption that the errors are normally distributed with homogeneity of variance.

Recall that the *errors* from the population model (Equation 1.5) differ from the *residuals* from the regression equation formed with the sample-based parameter estimates. Because the errors are unobservable, we must instead work with the residuals. With statistical software, it is possible to extract the residuals from a fitted regression model and then the residuals can be examined like any other variable. Returning to the movie example of *profit* regressed on *review*, Figure 1.5 presents a *quantile-quantile plot*, or **Q-Q plot**, of the residuals from this model against the quantiles of a normal distribution. Because many of the datapoints in this plot stray from the diagonal, solid reference line, this plot suggests that the model's residuals do not align with the expected normal distribution, hence providing evidence against the assumption of normally distributed model errors. Yet, because the sample size ( $N = 609$ ) is large, this result is not a serious concern.

[Figure 1.5 here]

### Homogeneity of variance

Perhaps more important than determining whether errors are normally distributed is assessing whether they are **homoscedastic**, meaning that their variance is consistent across the values of the regressor  $X_i$ ; in other words, it's important to examine the observed residuals for *homogeneity of variance*. In the movie example, because the regressor (*review*) is approximately continuous, it's best to examine the residuals conditioned on *review* using a scatterplot, as in Figure 1.6. To aid interpretation, this scatterplot is enhanced with a non-parametric LOWESS curve along with a smoother applied to the root-mean-square positive and negative residuals from the LOWESS curve (see Fox & Weisberg, 2019, pp. 136-137), which give a graphical summary of the spread of the data in the plot. Because these dashed curves become increasingly separated from each other as they move from low to high *review* scores, the residuals are evidencing *heteroscedasticity*, or heterogenous variance; that is, the spread of the residuals increases across the values of the regressor. Therefore, the homogeneity of variance assumption appears violated for this linear model. Again, however, the large sample size provides considerable robustness against this assumption violation.

[Figure 1.6 here]

[start box]

### Section recap: Basic concepts for regression diagnostics

**Regression diagnostics** are graphical and numeric procedures used to check the extent to which an estimated model's assumptions have been violated and to examine whether any unusual or outlying observations may be impacting the results.

The OLS linear regression assumptions of linear association, normally distributed errors, and homogeneity of error variance can be effectively assessed using plots of the *residuals* from an estimated regression model.

[end box]

### Examining outliers

In regression, as with most statistical procedures, it is important to be concerned about *influential observations* which can dramatically distort results. Datapoints that appear as outliers in simple scatterplots are often overly influential, but sometimes they are not, and sometimes such datapoints may even be considered “good” outliers. Therefore, in addition to examining data with basic scatterplots, it can also be valuable to use more sophisticated methods for determining whether any unusual cases are a cause for concern about the results of a regression analysis. To this end, a variety of statistical measures has been developed to assess the extent to which each case in a regression analysis may be considered influential. In general, a case or observation’s **influence** is a function of its **leverage** and **discrepancy**.

*Leverage* for an individual case is the extent to which that case differs from the rest of the cases in terms of scores on the regressor variable(s) only. For a simple regression model with a sole predictor  $X_i$ , the sample-based leverage of a given case  $i$ , which is also known as its **hat value**, is defined as

$$(1.8) \quad h_i = \frac{1}{N} + \frac{(X_i - \bar{X})^2}{\sum_{i=1}^N (X_i - \bar{X})^2}.$$

Equation 1.8 shows that leverage for case  $i$  is a function of how much the case deviates from the mean of  $X$  relative to the overall variation in  $X$  across the whole sample. Leverage ranges from 0 to 1, with larger values indicative of greater leverage.

*Discrepancy* is the extent to which a case has an extreme value of  $Y$  relative to its predicted value  $\hat{Y}_i$ ; discrepancy is thus the magnitude of the residual for case  $i$ . Discrepancy is usually assessed using a standardization of the residuals to facilitate the comparison of a given case with the rest of the sample. A common discrepancy measure is the **Studentized residual**, which helps determine whether a given residual is extreme relative to the other residuals. Specifically, the *externally* Studentized residual  $e_i^*$  (also known as the “Studentized deleted residual” or the “Jackknife residual”) for case  $i$  is its residual divided by a function of the standard deviation of the residuals from a regression equation estimated using all observations *except* for case  $i$ :

$$(1.9) \quad e_i^* = \frac{e_i}{s_{e(-i)} \sqrt{1 - h_i}},$$

where  $e_i = Y_i - \hat{Y}_i$  is the original, raw-scale residual for case  $i$ ,  $s_{e(-i)}$  is the standard deviation of the residuals from the regression model estimated after removing case  $i$  from the data, and  $h_i$  is the leverage for case  $i$ . The logic of removing case  $i$  for getting the  $s_{e(-i)}$  term in Equation 1.9 is that if this case is an outlier with respect to an estimated regression model, it can inflate the standard deviation of the residuals from that model. Thus, to measure whether a case’s residual is extreme relative to the other residuals, that case should be removed before calculating the standard deviation of the residuals of the other cases from a model estimated without including the potentially extreme case  $i$ .

An externally Studentized residual follows a  $t$ -distribution with  $N - 2 - P$  degrees of freedom (where  $P$  is the number of regressors in the model, with  $P = 1$  for simple regression).

Yet, although a Studentized residual is distributed as a  $t$ -statistic, it should not be evaluated against the usual  $t$ -distribution critical values because of the multiple comparison problem (i.e., there are many potential significance tests because there are many cases whose Studentized residuals can be calculated). Nonetheless, the  $t$ -distribution allows a rough guideline whereby externally Studentized residuals greater than 2.5 or so certainly can be considered extreme and potentially problematic.

If a case has extreme leverage or discrepancy, it may not actually have an excessive impact on the actual estimates of the parameters obtained from the data. But **influence** combines leverage and discrepancy to represent the extent to which a case affects a model's parameter estimates. One common influence measure is **Cook's distance**, defined as

$$(1.10) \quad D_i = \frac{e_i^*}{p+1} \times \frac{h_i}{1-h_i},$$

where  $P$  is again the number of regressors in the model. A large value of  $D_i$  indicates that case  $i$  has substantial influence on the parameter estimates. From Equation 1.10, it is apparent that Cook's  $D$  is a function of both leverage (i.e.,  $h_i$ ) and discrepancy (i.e.,  $e_i^*$ ), and that  $D_i$  will be large when both  $h_i$  and  $e_i^*$  are large. But a small  $h_i$  can compensate for a large  $e_i^*$  (and vice versa) to produce a  $D_i$  that is not particularly large; that is, a case can have large leverage or large discrepancy without having a large influence.

If an outlying case has a large influence, then the slope estimate  $\beta_1$  will change substantially if the model is re-estimated after removing that case from the data. Conversely, if a case has large discrepancy but low leverage and therefore low influence, then removing that case and re-estimating the model won't change  $\beta_1$  dramatically, but the standard error of  $\beta_1$  will become smaller, leading to a narrower confidence interval around  $\beta_1$  and greater power for testing the significance of  $\beta_1$ . Finally, a so-called "good leverage case" is an outlier large leverage but small discrepancy and thus small influence; the presence of such a case increases the precision of  $\beta_1$ ; removing a good leverage case is therefore counterproductive because doing so would increase the standard error of  $\beta_1$ . The online supplemental material provides a graphical demonstration of how outliers with differing combinations of leverage, discrepancy, and influence affect the basic results for a simple regression model.

The question of whether to remove outlying cases from a dataset is often difficult to answer. If it can be determined that a case has unusual values because of researcher error (e.g., a data entry error) or participant error (e.g., a research participant misunderstood instructions), then the case should be removed. In other situations, discovering an outlying case can be illuminating: Perhaps the outlying case was sampled from a sub-population that differs in important ways from the main population of interest. Otherwise, outliers represent legitimate data and researchers should not remove them solely for the purpose of "improving" results; outlier removal can be considered a "questionable research practice" that hinders the replicability of results. It is critical that researchers be transparent about the treatment of outliers in an analysis; a common practice is to report results obtained both with and without outlying cases included.

[start box]

### Section recap: Examining outliers

After a linear regression model has been fitted to sample data, the **leverage**, **discrepancy**, and **influence** of *each* case in the sample can be assessed.

Leverage, measured with *hat values*, represents how much a case is extreme relative to the distribution of the regressor variable(s) only.

Discrepancy, measured with *Studentized residuals*, represents how much the observed outcome value for a case is extreme relative to its predicted value.

Influence, measured with *Cook's Distance*, represents how much the leverage and discrepancy of a case combine to affect the model's parameter estimates.

If a case is an outlier with respect to leverage or discrepancy, it still might not have undue influence on the estimates of the regression parameters.

[end box]

#### A data-based illustration of leverage, discrepancy, and influence

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### Chapter Summary

- The simple linear regression model for the relation between an outcome variable  $Y$  and a regressor  $X$  is  $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ .
  - The **intercept parameter**,  $\beta_0$ , is the predicted value of  $Y$  when  $X$  equals 0.
  - The **slope parameter**,  $\beta_1$ , is the predicted difference in  $Y$  corresponding to a one-unit difference in  $X$ . The magnitude of  $\beta_1$  is therefore a (population) effect size for the association between  $X$  and  $Y$ .
- A scatterplot of an outcome variable against a potential regressor helps determine whether it is appropriate to model the relation between the two variables using a straight-line function, i.e., a linear regression equation.
- **Covariance** is a descriptive statistic measuring the strength of linear association between two variables; the **product-moment correlation** is a standardization of covariance that ranges between -1 and +1.
- A **predicted value**,  $\hat{Y}_i$ , is the outcome variable score for individual  $i$  that is predicted from the regression line given that individual's score on the regressor,  $X_i$ , whereas an **observed value**,  $Y_i$ , is the actual outcome variable score for individual  $i$  regardless of that individual's score on  $X_i$ .
  - When  $\hat{Y}_i$  is based on a population regression equation, the regression **error** for case  $i$  is  $\varepsilon_i = Y_i - \hat{Y}_i$ .
  - When  $\hat{Y}_i$  is based on an estimated regression equation, the **residual** for case  $i$  is  $e_i = Y_i - \hat{Y}_i$ .
- Model **specification** establishes a model's parameters using one or more equations giving the hypothetical mathematical relations among the variables. Often specification also involves statements about a model's assumptions regarding variance terms or probability distributions.



- Model **estimation** is the procedure by which the model's parameters are estimated from sample data.
  - Ordinary least squares (OLS) is the most common estimation method for linear regression. OLS produces parameter estimates that minimize the sum of squared residuals.

- The null hypothesis that  $\beta_1 = 0$  can be evaluated using a  $t$ -test with

$$t = \frac{\beta_1}{s_{\beta_1}}$$

where  $s_{\beta_1}$  is the estimated standard error of  $\beta_1$ .

- Even more information about the true parameter value is provided by a confidence interval, which can be calculated with

$$\beta_1 \pm s_{\beta_1} \times t_\alpha$$

where  $t_\alpha$  is the *critical t-value* for a  $(1 - \alpha)\%$  confidence interval.

- The validity of these inferential methods depends on the assumption that  $\varepsilon_i$  is normally distributed with constant (homogenous) variance across the range of  $X$ .
- When an outcome variable is regressed on a binary dummy variable, the intercept estimate equals the mean of the category (or group) coded zero,  $\beta_0 = Y_{X=0}$ , and the slope estimate equals the difference between the means of the two categories (or groups),  $\beta_1 = (Y_{X=1} - Y_{X=0})$ .
- **Regression diagnostics** are graphical and numeric procedures for checking the extent to which an estimated model's assumptions have been violated and to examine how any outlying observations may be impacting the results.
- Leverage, measured with *hat values*, represents how much a case is extreme relative to the distribution of the regressor variable(s) only.
- Discrepancy, measured with *Studentized residuals*, represents how much the observed outcome variable score for a case is extreme relative to its predicted value.
- Influence, measured with *Cook's Distance*, represents how much the leverage and discrepancy of a case combine to affect the model's parameter estimates.